FSSAI 0°C – 4°C CERTIFIED ACTIVE FRESHCHAIN

GOMEAT

Full-Stack Production MERN E-Commerce Architecture

A comprehensive engineering presentation covering bank-grade Razorpay payment rails, circular ₹250 refundable cold-box logistics, decoupled order lifecycle engines, 6-stage IoT telemetry, and Central Hub operations.

ARCHITECTURE
MERN Enterprise Stack
React 19 • Express • Node.js • MongoDB Atlas

PAYMENT RAILS
Razorpay Standard Suite
UPI (GPay, PhonePe, Paytm, QR) • Cards • NetBanking

CIRCULAR PACKAGING
ThermalVault™ System
₹250 Refundable Security Deposit Reversal

LIVE DEPLOYMENT
Vercel Serverless
CI/CD Production Deployment via GitHub

THE MARKET OPPORTUNITY

Industry Challenges & The GoMeat Solution

01

The Frozen Meat Illusion

MARKET DEFECT:
Standard delivery apps deliver rock-frozen cuts with broken cold-chains. Ice crystallization punctures meat cell membranes, leaching vital juices and flavor upon defrosting.

GOMEAT SOLUTION:
0°C to 4°C active chilled cold chain dispatched from micro-hubs in under 90 minutes. Fresh cut daily, never frozen.

02

Single-Use Styrofoam Waste

ENVIRONMENTAL CRISIS:
Every conventional meat delivery generates 1–2 kg of toxic thermocol (EPS) packaging and plastic bubble wrap that clogs municipal landfills for over 500 years.

GOMEAT SOLUTION:
ThermalVault™ reusable EPP insulated boxes with frozen gel packs. ₹250 refundable deposit credited back to UPI in 1 click upon box return.

03

Insecure & Tampered Checkouts

COMMERCE VULNERABILITY:
Most amateur stores trust frontend-submitted prices, use fake simulated payments, or static QR codes prone to user tampering and chargeback losses.

GOMEAT SOLUTION:
Tamper-proof server-side price calculation from MongoDB, HMAC SHA-256 signatures, and cryptographic raw-body webhook verification.

TECHNICAL ARCHITECTURE

Full-Stack Production System Architecture

CLIENT LAYER

Frontend Client

React 19 • Vite • Tailwind

- Responsive mobile-first e-commerce UI with Figma order flow
- Ambient radiant lighting mesh with Dark/Light theme toggle
- Client state persistence for Cart, Auth, and User Wishlist
- Dynamic script injection for Razorpay Standard Checkout SDK
- Non-destructive browser back navigation via popstate sync

`Bundle: 527 kB gzip (2.56s)`

API LAYER

Express REST Router

Node.js Express • CORS • JWT

- Raw request body retention for cryptographic Webhooks
- JWT Authentication & Role-Based Access Control (RBAC)
- /api/payments (create-order, verify, webhook, status, refund)
- /api/orders (pipeline status updater, customer cancellation)
- /api/products (full CRUD, live stock adjustments, filters)

`Vercel Serverless Function`

PAYMENT GATEWAY

Razorpay Engine

HMAC SHA-256 • Crypto API

- Server-side order creation calculated from DB product prices
- Cryptographic HMAC SHA-256 signature verification
- Automated duplicate payment & double-order idempotency
- Razorpay Refund API for ₹250 cold-box deposit reversals
- Multi-rail support: UPI, Cards, NetBanking, Wallets, and COD

`Paise precision (INR)`

PERSISTENCE

MongoDB Atlas

Replica Set • Mongoose 9

- Cloud hosted database cluster: cluster0.6rm7klj.mongodb.net
- Atomic stock decrement upon payment capture
- Order price snapshots & complete address hierarchy
- Independent decoupled orderStatus & paymentStatus
- Dual-engine persistence: Live MongoDB + zero-downtime cache

`sagarfresh production DB`

PAYMENT ENGINEERING

Module 1: Real Razorpay Payment Gateway Integration

ENTERPRISE PAYMENT REQUIREMENTS MET

1. Server-Side Price Calculation:
Prices are calculated strictly by querying MongoDB product records. Frontend submitted totals are discarded to prevent inspect-element tampering.

2. HMAC SHA-256 Signature Verification:
Crypto engine computes HMAC SHA-256 over (razorpay_order_id + '|' + razorpay_payment_id) with RAZORPAY_KEY_SECRET to guarantee authentic bank confirmation.

3. Raw Body Webhook Verification:
Preserved express raw body for raw byte-matching against x-razorpay-signature, supporting captured, failed, and refund events.

4. Idempotency & Duplicate Prevention:
Re-verifying an already PAID order returns 200 OK immediately with the existing order, preventing duplicate records or multiple debits.

CRYPTOGRAPHIC PAYMENT FLOW

1. **POST /api/payments/create-order**
   Backend looks up cuts in DB, validates stock, adds ₹250 deposit, and calls Razorpay SDK to generate order_id in paise.
2. **Razorpay Standard Checkout Modal**
   Dynamic checkout.js launches official Razorpay modal. User completes payment via GPay, PhonePe, Cards, or NetBanking.
3. **POST /api/payments/verify**
   Client submits razorpay_order_id, razorpay_payment_id, and signature. Node.js crypto engine authenticates HMAC SHA-256 signature.
4. **Atomic Persistence & Stock Deduction**
   Decrements product inventory in MongoDB, locks cold-box asset serial number, and transitions orderStatus to CONFIRMED.

```
Verified Test Order: GOM-280363 • Status: PAID • HMAC-SHA256 Match OK
```

CHECKOUT & ORDER ENGINEERING

Module 2 & 3: Complete Checkout & Decoupled Order Lifecycle

MODULE 2: COMPLETE CHECKOUT ARCHITECTURE

1. Customer Contact Details:
Full Name, Email Address (for automated tax invoices), and 10-digit Indian Mobile number validation.

2. Delivery Slot Picker:
Choose between "⚡ Express 90 Mins Delivery" (instant dark depot dispatch) or "🗓️ Scheduled Slots" (Today/Tomorrow).

3. Hierarchical Delivery Address:
Address type (Home/Work/Other), Flat/House No, Building/Society Name, Street, Locality, City, State, and 6-digit PIN.

4. Rider Instructions & Landmark Notes:
Quick tags: "Leave with security guard", "Do not ring bell (Sleeping baby)", plus custom delivery directions.

5. Transparent Order Summary:
Subtotal, separate ₹250 refundable cold-box deposit line item, coupon discounts, shipping fee, and payable total.

MODULE 3: DECOUPLED ORDER STATE ENGINE

Payment Status and Order Pipeline Status are independently managed to allow retries, webhooks, cancellations, and refunds without data corruption.

PENDING	Order created; awaiting bank authorization
CONFIRMED	**Payment verified; queued for butchery prep**
PREPARING	Master butchers trimming fresh cuts to spec
QUALITY_CHECK	FSSAI inspection, digital scale & 0°C probe
PACKED	Sealed in ThermalVault with sub-zero ice gels
OUT_FOR_DELIVERY	In transit with EV cold-chain fleet partner
DELIVERED	**Delivered chilled; ₹250 refundable deposit recorded**

Order Cancellation Guard: Customers can cancel anytime prior to PACKED stage to receive an automated refund and restock depot inventory.

CUSTOMER EXPERIENCE

Module 4: Customer Dashboard Architecture

SECTION 1

My Profile

Customer credentials, verified Google Sign-In badge, registered mobile number, delivery city, and active refundable deposit balance held.

1-Click Profile Update

SECTION 2

My Orders History

Historical log of all placed orders with unique order numbers (GOM-XXXX), item snapshots, timestamps, live pipeline tags, and 1-click reorder.

Real-time status tracking

SECTION 3

Order Details Modal

Inspect itemized price breakdowns, cut customization specifications, delivery rider contacts, and active cold-probe telemetry logs.

Price & Address Snapshot

SECTION 4

Payment History Ledger

Cryptographic ledger displaying Razorpay Order IDs, Payment IDs, payment methods (UPI/Cards/COD), and refund states for complete transparency.

Cryptographic audit trail

SECTION 5

Wishlist Management

Heart-tagged favorite seafood, mutton, and poultry cuts stored in persistent client memory with 1-click instant "Add to Cold Box".

Session persistence

SECTION 6

Saved Addresses

Address book managing multiple delivery locations (Home, Work, Other) with complete flat, society, landmark, and PIN details.

1-Tap Checkout Autofill

SECTION 7

Cold-Box Returns Portal

Track circulating ThermalVault boxes and request free doorstep collection with instant ₹250 UPI refund reversal upon courier pickup.

Zero-waste circular flow

SECTION 8

Printable Tax Invoices

1-click automated tax invoice generator showing FSSAI license, GSTIN, item breakdowns, and itemized ₹250 deposit for commercial records.

Print & PDF Preview

TELEMETRY & LOGISTICS

Module 5: Status-Based Tracking & IoT Cold Chain Telemetry

6-STAGE VISUAL MILESTONE PIPELINE

Status-Based (No Fake GPS)

1	**Order Confirmed** Central depot verifies payment and queues cuts for master butchery.	0-10 Mins
2	**Preparing Your Order** Master butchers precision-trim, bone, and portion cuts in bio-secure rooms.	10-25 Mins
3	**Quality Check** 150-point FSSAI inspection, digital weigh-in, and microbial assay check.	25-35 Mins
4	**Packed in Cold Box** Sealed inside ThermalVault with sub-zero ice gel packs; digital lock engaged.	35-50 Mins
5	**Out for Delivery** Assigned to EV cold-chain partner heading directly to delivery doorstep.	50-75 Mins
6	**Delivered Chilled** Handover verified; ₹250 refundable cold-box deposit credited to account.	Under 90 Mins

COLD CHAIN TELEMETRY

ThermalVault™ IoT Sensor Core

INTERNAL BOX PROBE TEMP

0.8°C - 2.3°C

FSSAI 0°C-4°C Safe FreshZone Verified

Packaging Status:	EPP Ice-Gel Seal Intact
Assigned Fleet Executive:	Ramesh Kumar (EV Fleet #12)
Digital Seal ID:	#FS-84920-BLR
Live Courier Contact:	1-Tap Direct Call & WhatsApp

Continuous tamper-evident probe logging prevents spoilage across all transit hops.

CIRCULAR PACKAGING ARCHITECTURE

Module 6: Signature ₹250 Refundable Cold-Box System

STAGE 1: CHECKOUT

Transparent Deposit Billing

A separate ₹250 refundable security deposit is itemized in the cart, included in the order summary, and clearly recorded on the GST/FSSAI tax invoice.

```
Items Subtotal + ₹250 Box Deposit
```

STAGE 2: DELIVERY

ThermalVault Handover

Customer receives fresh cuts in pristine sanitized EPP cold box. Customer can immediately return empty box to courier or retain it for doorstep collection.

```
Serial Tag: BOX-BLR-412
```

STAGE 3: RETURN REQUEST

1-Click Pickup Initiation

Customer opens Dashboard and clicks "Schedule Box Return". Customer specifies convenient pickup window and verifies destination UPI ID for refund.

```
State: REQUESTED
```

STAGE 4: HUB DEPOT AUDIT

Sanitization & Verification

Dark depot logistics team inspects the returned EPP box, cleans surfaces with food-grade bio-sanitizer, verifies gel bricks, and logs condition.

```
Inspection: EPP Walls & Bricks Intact
```

STAGE 5: EXECUTION

Razorpay Refund API Trigger

Admin triggers 1-click refund via POST /api/payments/refund-coldbox. Razorpay Refund API processes ₹250 reversal directly back to source payment.

```
API: POST /payments/refund-coldbox
```

STAGE 6: PROTECTION

Duplicate Refund Guard

System stores refundTxnId and transitions state to REFUNDED. Subsequent attempts return 400 Bad Request to strictly prevent double refunds.

```
State: REFUNDED (Idempotent Guard)
```

HUB FULFILLMENT OPS

Module 7: Central Hub Operations Admin Control Room

TOTAL REVENUE	TOTAL ORDERS	PENDING ORDERS	COMPLETED	TOTAL CUSTOMERS	VERIFIED PAYMENTS
₹1,48,500	142 Orders	24 Active	118 Orders	86 Profiles	100% SHA256
Razorpay & COD	All time volume	In prep / On road	Delivered safely	Registered users	HMAC Authenticated

1. ORDER PIPELINE & DISPATCH CONTROL

- Real-time search by Order ID, Customer Name, or Phone
- 7-stage dropdown updater (PENDING to DELIVERED)
- Instant status sync across customer tracking view
- Order cancellation with automatic inventory restocking
- Transparent view of cut custom styles and rider notes

2. INVENTORY MANAGEMENT & STOCK ALERTS

- Real-time automated low-stock warnings for items < 10 units
- Critical alert banner identifying impending stockouts
- Quick +/- inline stock adjuster (+5, +10, +50 bulk add)
- Add new fresh cuts with gross vs net weight transparency
- Mark cuts unavailable to prevent checkout failures

3. PAYMENT & COLD-BOX REFUND EXECUTION

- Complete Razorpay transaction ledger with payment IDs
- Separate tracking for circulating ₹250 cold-box deposits
- 1-click "Refund ₹250" trigger calling Razorpay API directly
- Automated idempotent refund logging with refundTxnId
- Fleet tracking: ThermalVault units available vs with customer

CUSTOMER EXPERIENCE ENGINEERING

Module 8 & 9: Premium UI/UX & Customer Experience

Multi-Criteria Sorting

Sort cuts by Featured, Price (Low to High), Price (High to Low), or Highest Discount % for rapid customer discovery.

Price Budget Filter Pills

One-tap budget filtering: All Prices, Under ₹300, Under ₹500, and Premium Feast cuts (₹500+).

Persistent Wishlist

Heart-toggle button on every cut card and detail modal; persists across sessions and syncs to customer dashboard.

Instant Toast Feedback

Non-intrusive floating toasts notify users of cart additions, wishlist updates, and background operations.

Ambient Lighting & Dark Mode

Lush dark and light themes with glowing radial mesh orbs, glassmorphism cards, and neon border transitions.

Custom Butchery Cuts

Product modal lets users select custom cuts: Curry Cut, Biryani Cut, Boneless Fillet, or Hand-Minced Keema.

Zero Login Barrier

Users explore cuts freely; Google OAuth or credentials only prompted at checkout to maximize conversion rates.

Non-Destructive Back Navigation

Browser history synchronization ensures pressing "Back" retreats by 1 step/modal instead of exiting the site.

VERIFICATION & QUALITY ASSURANCE

Module 10: Security Engineering & End-to-End Test Matrix

TEST SCENARIO	TEST METHODOLOGY & PARAMETERS	RESULT & OBSERVATION	VERIFICATION STATUS
Server-Side Price Calculation	Submitted fake prices and invalid prod_chicken_curry ID	Rejected untrusted price; calculated exact DB total (2× ₹289 + ₹250 = ₹868)	PASSED
HMAC SHA-256 Verification	Crypto hash verification on POST /api/payments/verify	Signature matched; transitioned order GOM-280363 to PAID & CONFIRMED	PASSED
Idempotency & Double-Charge	Re-called verification API on already PAID order	Returned 200 OK with cached order without creating duplicates	PASSED
Razorpay Cold-Box Refund	POST /api/payments/refund-coldbox on order GOM-280363	Generated refundId rfnd_df866178; status updated to REFUNDED	PASSED
Duplicate Refund Prevention	Attempted second refund call on same order	Blocked with 400 Bad Request: "₹250 deposit already refunded"	PASSED
Client TypeScript Build	Executed tsc -b && vite build across 2052 modules	Zero compiler errors; built production bundle in 2.56 seconds	PASSED
Live Vercel Health Check	Queried GET /api/health on production Vercel deployment	HTTP 200 OK {"status":"ok","name":"GoMeat API"} verified	PASSED

BUSINESS IMPACT & ROADMAP

A Truly Sustainable, Scalable Meat E-Commerce Platform

0 kg

Single-Use Styrofoam

100% circular packaging eliminates disposable thermocol boxes and plastic bubble wrap.

90 Mins

Hyperlocal Delivery

Active chilled 0°C–4°C cold chain dispatched directly from neighborhood micro-hub dark stores.

100%

Tamper-Proof Rails

Cryptographic SHA-256 signatures, server-side DB pricing, and bank-grade Razorpay checkout.

1-Click

Automated Refunds

Instant ₹250 cold-box deposit reversals executed directly through the Razorpay Refund API.

LIVE PRODUCTION LINKS

Web: https://gomeat-git-main-dhinesh-s-projects1.vercel.app • API: /api/health (200 OK)

Repository: Dhinesh1628/sagarfresh (main)